assuming we are currently @ time t

$$\boxed{L_{t+1}} = L_t + \alpha (y_t - L_t)$$

L_{t+1}: one-step forecast ⇓ f_{t+1}, $L_{t+1|t}$

L_t: forecast for my current period (performed at t-1)

α: smoothing parameter $0<\alpha<1$

$(y_t - L_t)$: error in my forecast for the current period

Q: how to choose α?

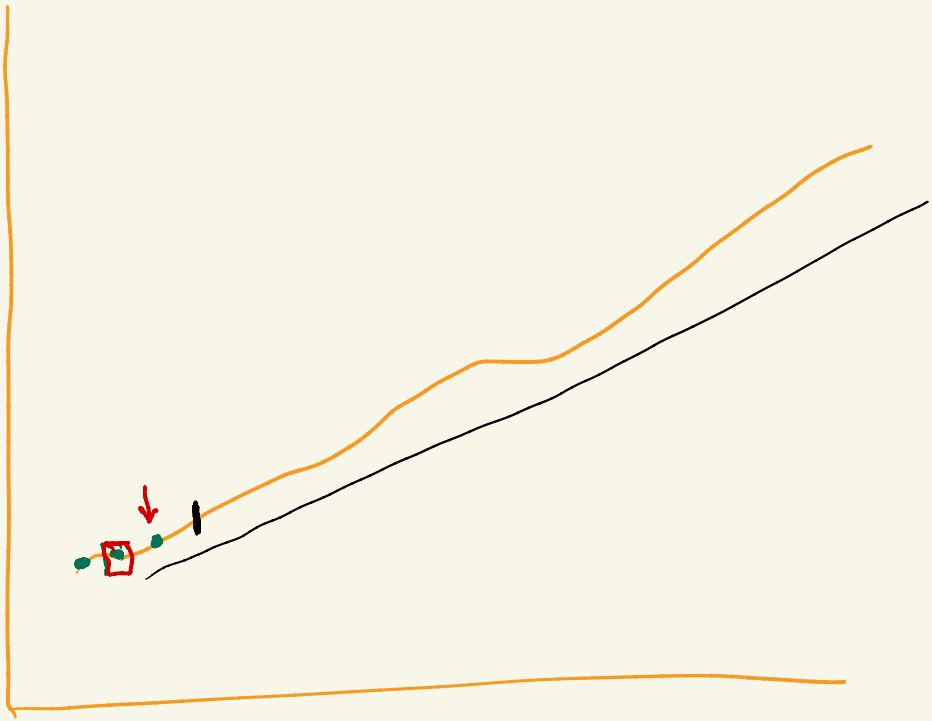
- heuristic ⇓ choose a value of α that works well in similar problems e.g. $\alpha = 0.2$
- figure out the best value for α from a subset of common α values
- $\boxed{\text{optimize}}$ α based on a training sample

$$L_{t+1} = \alpha y_t + (1-\alpha) L_t$$

α → smoothing par

↳ interpret weight given for the current value of ts

$(1-\alpha)$ → weight given to my forecast



while the SES approach will result in poor forecasts, which α would you recommend?

$\alpha = 0.2$

$\alpha = 0.5$

$\alpha = 0.9$